Unit 4

# The d- and f-Block Elements

Transition elements such as iron, copper, silver and gold have shaped the growth of human civilisation, and inner transition elements such as Th, Pa and U are now proving to be excellent sources of nuclear energy.

## What this unit sets out to do

- find where the d- and f-block elements sit in the periodic table
- learn how electrons are arranged in transition (d-block) and inner transition (f-block) atoms
- judge the relative stability of oxidation states from electrode potential values
- describe how important compounds such as $K_2Cr_2O_7$ and $KMnO_4$ are made, and their properties, structures and uses
- understand the general features of d- and f-block elements and their horizontal and group trends
- describe the f-block elements and compare lanthanoids with actinoids in electronic configuration, oxidation states and chemical behaviour

## The two blocks

- **d-block**: groups 3–12, where d orbitals fill step by step across each of the four long periods
- **f-block**: elements filling 4f and 5f orbitals, set in a separate panel at the foot of the periodic table
- the d- and f-block elements are often called **transition metals** and **inner transition metals**

## The series

- transition metals: 3d (Sc to Zn), 4d (Y to Cd), 5d (La and Hf to Hg), and 6d (Ac and Rf to Cn)
- inner transition metals: 4f (Ce to Lu), the **lanthanoids**; 5f (Th to Lr), the **actinoids**

## What "transition" means

- the name first arose because their chemistry was transitional between that of s- and p-block elements
- IUPAC now defines transition metals as metals with an incomplete d subshell in the neutral atom or in their ions

## Zn, Cd and Hg

- group 12 metals have a full $d^{10}$ configuration in the ground state and in their common oxidation states, so are not counted as transition metals
- as the end members of the 3d, 4d and 5d series, their chemistry is studied alongside the transition metals
- partly filled d or f orbitals set transition elements apart from the non-transition elements

### Studied on their own

- so transition elements and their compounds are treated separately
- yet the ordinary valence theory used for non-transition elements also works well for them
- precious metals (silver, gold, platinum) and industrially important metals (iron, copper, titanium) are transition metals

### Plan of the unit

- first: electronic configuration, occurrence and general features of transition elements, stressing trends in the first row (3d), plus preparation and properties of some key compounds
- then, for the inner transition metals: how their electrons are arranged, which oxidation states they show, and how reactive they are

# The Transition Elements (d-Block)

## 4.1 Position in the Periodic Table

- the d-block fills the broad centre of the periodic table, between the s- and p-blocks
- electrons enter the d orbitals of the penultimate energy level, giving four rows of transition metals: 3d, 4d, 5d and 6d (Table 4.A)

## 4.2 Electronic Configurations of the d-Block Elements

### General configuration

$$(n-1)d^{1-10}\,ns^{1-2}$$

- exception: Pd is $4d^{10}5s^{0}$
- $(n-1)$ marks the inner d orbitals, holding one to ten electrons; the outermost $n$s orbital holds one or two
- there are several exceptions, because $(n-1)$d and $n$s differ very little in energy, and half-filled and completely filled sets are especially stable

### Cr and Cu

- **Cr**: $3d^{5}\,4s^{1}$, not $3d^{4}4s^{2}$; the 3d–4s energy gap is small and the half-filled 3d set is favoured
- **Cu**: $3d^{10}4s^{1}$, not $3d^{9}4s^{2}$
- ground-state outer configurations of all transition elements: Table 4.A

TABLE 4.A Electronic Configurations of outer orbitals of the Transition Elements (ground state)

| | 1st Series | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | **Sc** | **Ti** | **V** | **Cr** | **Mn** | **Fe** | **Co** | **Ni** | **Cu** | **Zn** |
| *Z* | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 4s | 2 | 2 | 2 | 1 | 2 | 2 | 2 | 2 | 1 | 2 |
| 3d | 1 | 2 | 3 | 5 | 5 | 6 | 7 | 8 | 10 | 10 |

Table 4.A (continued)

| 2nd Series | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Y | Zr | Nb | Mo | Tc | Ru | Rh | Pd | Ag | Cd |
| *Z* | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 |
| 5s | 2 | 2 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 2 |
| 4d | 1 | 2 | 4 | 5 | 6 | 7 | 8 | 10 | 10 | 10 |
| **3rd Series** | | | | | | | | | | |
| | La | Hf | Ta | W | Re | Os | Ir | Pt | Au | Hg |
| *Z* | 57 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 6s | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 2 |
| 5d | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 9 | 10 | 10 |
| **4th Series** | | | | | | | | | | |
| | Ac | Rf | Db | Sg | Bh | Hs | Mt | Ds | Rg | Cn |
| *Z* | 89 | 104 | 105 | 106 | 107 | 108 | 109 | 110 | 111 | 112 |
| 7s | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 2 |
| 6d | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 10 |

### Zn, Cd, Hg and Cn

- outer configuration $(n–1)d^{10}ns^2$
- their orbitals are completely filled in the ground state and in their common oxidation states, so they are not regarded as transition elements

### Why d orbitals matter

- d orbitals reach out to the atom's edge more than s and p orbitals, so they are more affected by, and affect more, the surrounding atoms or molecules
- ions with the same $d^n$ configuration ($n$ = 1–9) share some magnetic and electronic properties

### Features from partly filled d orbitals

- a variety of oxidation states; coloured ions; complex formation with many ligands
- catalytic properties and paramagnetism in the metals and their compounds (all treated in detail later)

### Rows over groups

- transition elements resemble each other more along a horizontal row than non-transition elements do, though some group likenesses exist
- row trends (especially the 3d row) come first, then group likenesses

### Worked example scandium vs zinc

Why is scandium ($Z$ = 21) a transition element while zinc ($Z$ = 30) is not?

- ground-state scandium has an incompletely filled 3d set ($3d^1$), so it counts as a transition element
- zinc's d orbitals are completely filled ($3d^{10}$) both in the ground state and in its oxidised state, so it does not

### Intext question

**Q1** In its ground state a silver atom has completely filled d orbitals ($4d^{10}$). On what basis is it still called a transition element?

## 4.3 General Properties of the Transition Elements (d-Block)

The sections that follow deal only with the elements of the first transition series.

## 4.4 Physical Properties

### Typically metallic

- nearly all are strong in tension, ductile and malleable, conduct heat and electricity well, and have a metallic shine
- apart from Zn, Cd, Hg and Mn, each has one or more typical metallic structures at ordinary temperatures (Table 4.B)

### Hard, with high melting points

- except Zn, Cd and Hg they are very hard, of low volatility, with high melting and boiling points (Figure 4.A)
- the high melting points come from more electrons, from (*n*–1)d as well as *n*s, taking part in interatomic metallic bonding
- along a row, melting points climb to a maximum at $d^5$ (Mn and Tc are anomalous) and then fall steadily with rising atomic number

TABLE 4.B Lattice Structures of Transition Metals

| Sc | Ti | V | Cr | Mn | Fe | Co | Ni | Cu | Zn |
|---|---|---|---|---|---|---|---|---|---|
| hcp (bcc) | hcp (bcc) | bcc | bcc | X (bcc, ccp) | bcc (hcp) | ccp (hcp) | ccp | ccp | X (hcp) |
| **Y** | **Zr** | **Nb** | **Mo** | **Tc** | **Ru** | **Rh** | **Pd** | **Ag** | **Cd** |
| hcp (bcc) | hcp (bcc) | bcc | bcc | hcp | hcp | ccp | ccp | ccp | X (hcp) |
| **La** | **Hf** | **Ta** | **W** | **Re** | **Os** | **Ir** | **Pt** | **Au** | **Hg** |
| hcp (ccp, bcc) | hcp (bcc) | bcc | bcc | hcp | hcp | ccp | ccp | ccp | X |

bcc = body centred cubic; hcp = hexagonal close packed; ccp = cubic close packed; X = a typical metal structure

**Figure 4.A** Trends in melting points of transition elements

- melting point (in $10^3$ K, roughly 1 to 3.6) against atomic number, with separate lines for the 3d, 4d and 5d series
- 3d (Ti … Cu): rises to Cr, dips sharply at Mn, then falls towards Cu; 4d (Zr … Ag): peaks at Mo, dips at Tc; 5d (Hf … Au): the highest line, peaking at W, then falling steeply to Au
- the heavier series lie above the 3d series throughout most of the range

### Enthalpies of atomisation

- transition metals have high enthalpies of atomisation (Figure 4.B)
- the maxima near the middle of each series show that one unpaired electron per d orbital particularly favours strong interatomic interaction
- in general, more valence electrons means stronger bonding

### Links to reactivity

- enthalpy of atomisation is a key factor in a metal's standard electrode potential; metals with very high values (very high boiling points) tend to be noble in their reactions (see electrode potentials later)
- second- and third-series metals have larger enthalpies of atomisation than their first-series partners, which helps explain the far more frequent metal–metal bonding in compounds of the heavy transition metals

**Figure 4.B** Trends in enthalpies of atomisation of transition elements

- $\Delta_a H^{\ominus}$ (kJ mol$^{-1}$, 0 to about 900) against atomic number for Series 1, 2 and 3
- each series rises to a peak in the middle and falls away at the end; Series 1 has a pronounced dip in the middle of its range
- Series 3 lies highest, Series 2 next, and Series 1 lowest over most of the range

## 4.5 Variation in Atomic and Ionic Sizes of Transition Metals

### Across a series

- ions of equal charge in a series shrink steadily as atomic number increases
- each unit rise in nuclear charge adds an electron to a d orbital, and d electrons shield poorly, so the pull of the nucleus on the outermost electron grows and the radius falls
- atomic radii follow the same trend, though the change within a series is quite small

### Between series

- radii increase from the first (3d) to the second (4d) series, but the third (5d) series has almost the same radii as the second (Figure 4.C)
- the cause: the 4f orbitals must fill before the 5d series starts
- filling 4f before 5d produces a steady fall in atomic radii, the **lanthanoid contraction**, which essentially cancels the size increase expected with rising atomic number

### Result of the lanthanoid contraction

- the second and third d series have similar radii (e.g. Zr 160 pm, Hf 159 pm)
- their physical and chemical properties are far more alike than ordinary family relationships would suggest

### Its cause

- similar to that in an ordinary transition series: imperfect shielding of one electron by another in the same set of orbitals
- but one 4f electron shields another even less than one d electron shields another; as nuclear charge rises along the series, the whole $4f^n$ set shrinks fairly regularly

### Density

- shrinking metallic radius together with rising atomic mass makes density increase in general
- note the marked rise from titanium ($Z$ = 22) to copper ($Z$ = 29) in Table 4.C

**Figure 4.C** Trends in atomic radii of transition elements

- radius (axis marked 12 to 19) for the 3d, 4d and 5d series: each falls from its first member towards the middle, then levels off or rises slightly at the end
- the 4d and 5d curves nearly coincide, both above the 3d curve

TABLE 4.C **Electronic Configurations and some other Properties of the First Series of Transition Elements**

| Element | Sc | Ti | V | Cr | Mn | Fe | Co | Ni | Cu | Zn |
|---|---|---|---|---|---|---|---|---|---|---|
| Atomic number | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Config. M | $3d^14s^2$ | $3d^24s^2$ | $3d^34s^2$ | $3d^54s^1$ | $3d^54s^2$ | $3d^64s^2$ | $3d^74s^2$ | $3d^84s^2$ | $3d^{10}4s^1$ | $3d^{10}4s^2$ |
| $M^+$ | $3d^14s^1$ | $3d^24s^1$ | $3d^34s^1$ | $3d^5$ | $3d^54s^1$ | $3d^64s^1$ | $3d^74s^1$ | $3d^84s^1$ | $3d^{10}$ | $3d^{10}4s^1$ |
| $M^{2+}$ | $3d^1$ | $3d^2$ | $3d^3$ | $3d^4$ | $3d^5$ | $3d^6$ | $3d^7$ | $3d^8$ | $3d^9$ | $3d^{10}$ |
| $M^{3+}$ | [Ar] | $3d^1$ | $3d^2$ | $3d^3$ | $3d^4$ | $3d^5$ | $3d^6$ | $3d^7$ | – | – |
| $\Delta_aH^\ominus$ / kJ mol$^{-1}$ | 326 | 473 | 515 | 397 | 281 | 416 | 425 | 430 | 339 | 126 |
| $\Delta_iH^\ominus_I$ / kJ mol$^{-1}$ | 631 | 656 | 650 | 653 | 717 | 762 | 758 | 736 | 745 | 906 |
| $\Delta_iH^\ominus_{II}$ | 1235 | 1309 | 1414 | 1592 | 1509 | 1561 | 1644 | 1752 | 1958 | 1734 |
| $\Delta_iH^\ominus_{III}$ | 2393 | 2657 | 2833 | 2990 | 3260 | 2962 | 3243 | 3402 | 3556 | 3837 |
| Radius / pm: M | 164 | 147 | 135 | 129 | 137 | 126 | 125 | 125 | 128 | 137 |
| $M^{2+}$ | – | – | 79 | 82 | 82 | 77 | 74 | 70 | 73 | 75 |
| $M^{3+}$ | 73 | 67 | 64 | 62 | 65 | 65 | 61 | 60 | – | – |
| $E^\ominus$/V: $M^{2+}$/M | – | –1.63 | –1.18 | –0.90 | –1.18 | –0.44 | –0.28 | –0.25 | +0.34 | –0.76 |
| $M^{3+}/M^{2+}$ | – | –0.37 | –0.26 | –0.41 | +1.57 | +0.77 | +1.97 | – | – | – |
| Density / g cm$^{-3}$ | 3.43 | 4.1 | 6.07 | 7.19 | 7.21 | 7.8 | 8.7 | 8.9 | 8.9 | 7.1 |

### *Worked example* enthalpy of atomisation

**Why are the enthalpies of atomisation of transition elements high?**

- their atoms carry many unpaired electrons
- this gives stronger interatomic interaction, hence stronger bonding between atoms and higher enthalpies of atomisation

### *Intext question*

**Q2** Across Sc ($Z = 21$) to Zn ($Z = 30$), zinc has the lowest enthalpy of atomisation, 126 kJ $mol^{-1}$. Explain why.

## 4.6 Ionisation Enthalpies

### *Overall trend*

- ionisation enthalpy rises from left to right along each series, as nuclear charge grows while the inner d orbitals fill
- Table 4.C gives the first three ionisation enthalpies for the first series; successive values rise less steeply than for non-transition elements
- values change far less along a transition series than along a period of non-transition elements
- while the first ionisation enthalpy mostly creeps upward, the second and third jump by far bigger amounts from element to element

### *Irregular first ionisation enthalpy (3d)*

- of little chemical importance; explained by removal of one electron changing the relative energies of 4s and 3d
- d-block elements lose *n*s electrons before (*n*–1)d electrons on forming ions
- from Sc to Zn nuclear charge rises, but electrons enter the inner 3d subshell; 3d electrons shield the 4s electrons from the growing nuclear charge somewhat better than outer-shell electrons shield each other
- so atomic radii shrink less rapidly and ionisation energies rise only slightly across the series

### *Second and third ionisation*

- doubly or more highly charged ions have $d^n$ configurations with no 4s electrons
- the second ionisation enthalpy is expected to rise generally with effective nuclear charge, since d orbitals point in different directions and one d electron does not shield another from the nucleus
- the steady rise breaks at $Mn^{2+}$ (second) and $Fe^{3+}$ (third), both $d^5$ ions; similar breaks occur at the matching elements of later series

### *Explaining $d^n$ variation*

- three terms fix the value of ionisation enthalpy:
- the attraction of each electron to the nucleus
- the repulsion between electrons
- the exchange energy

### Exchange energy

- exchange energy stabilises an energy state; it is roughly proportional to the total number of possible pairs of parallel spins in degenerate orbitals
- with several electrons in a degenerate set, the lowest-energy state has as many singly occupied orbitals with parallel spins as possible (Hund's rule)
- the greater the exchange-energy stabilisation that ionisation would destroy, the more stable the configuration and the harder the ionisation; at $d^6$ there is no such loss

### Consequences

- $Mn^+$ is $3d^5 4s^1$ while $Cr^+$ is $d^5$, so $Mn^+$ ionises more easily than $Cr^+$
- $Fe^{2+}$ is $d^6$ and $Mn^{2+}$ is $3d^5$, so $Fe^{2+}$ ionises more easily than $Mn^{2+}$: the third ionisation enthalpy of Fe is lower than that of Mn

### Forming $M^{2+}$

- +2 is the lowest common oxidation state; making gaseous $M^{2+}$ needs the enthalpy of atomisation plus the first and second ionisation enthalpies
- the second ionisation enthalpy dominates; it is unusually high for Cr and Cu, whose $M^+$ ions are $d^5$ and $d^{10}$
- for Zn it is correspondingly low, since removing one 4s electron leaves a stable $d^{10}$ configuration

### Third ionisation enthalpy

- not complicated by the 4s factor; shows how hard it is to remove an electron from $d^5$ ($Mn^{2+}$) and $d^{10}$ ($Zn^{2+}$) ions
- generally quite high; the high values for Cu, Ni and Zn explain why oxidation states above two are hard to reach for them
- ionisation enthalpies give some guide to the relative stability of oxidation states, but the question is very complex and resists easy generalisation

## 4.7 Oxidation States

- a striking feature of transition elements is the wide variety of oxidation states they show in compounds
- Table 4.D sets out the usual oxidation states of the first-row metals

TABLE 4.D **Oxidation States of the first row Transition Metal (the most common ones are in bold types)**

| Sc | Ti | V | Cr | Mn | Fe | Co | Ni | Cu | Zn |
|---|---|---|---|---|---|---|---|---|---|
| | +2 | +2 | +2 | **+2** | **+2** | **+2** | **+2** | +1 | **+2** |
| +3 | +3 | +3 | **+3** | +3 | **+3** | **+3** | +3 | **+2** | |
| | **+4** | +4 | +4 | +4 | +4 | +4 | +4 | | |
| | | **+5** | +5 | +5 | | | | | |
| | | | **+6** | +6 | +6 | | | | |
| | | | | **+7** | | | | | |

### Most states in mid-series

- the elements with the most oxidation states lie in or near the middle of the series; manganese shows every state from +2 to +7
- fewer states at the ends: too few electrons to lose or share (Sc, Ti), or too many d electrons, leaving fewer orbitals for sharing at higher valence (Cu, Zn)
- so scandium(II) is virtually unknown and Ti(IV) is more stable than Ti(III) or Ti(II); zinc shows only +2 (no d electrons involved)

### Highest stable states

- up to manganese, the highest reasonably stable oxidation state equals the sum of s and d electrons: $Ti^{IV}O_2$, $V^{V}O_2^{+}$, $Cr^{VI}O_4^{2-}$, $Mn^{VII}O_4^{-}$
- after that, higher states lose stability quite abruptly; typical species are $Fe^{II,III}$, $Co^{II,III}$, $Ni^{II}$, $Cu^{I,II}$, $Zn^{II}$

### Steps of one

- variable oxidation states come from incomplete filling of d orbitals, so they differ from each other by one: $V^{II}$, $V^{III}$, $V^{IV}$, $V^{V}$
- non-transition elements, by contrast, usually show states differing by two

### Trend down a group

- in groups 4 to 10 the heavier members favour **higher** oxidation states, the reverse of the p-block, where the inert pair effect favours lower states in heavier members
- in group 6, Mo(VI) and W(VI) are more stable than Cr(VI); acidic dichromate, Cr(VI), oxidises strongly, whereas $MoO_3$ and $WO_3$ do not

### Low oxidation states

- found in complexes whose ligands can act as π-acceptors as well as σ-bonding
- in $Ni(CO)_4$ and $Fe(CO)_5$, nickel and iron are in oxidation state zero

### Worked example no variable oxidation state

**Name a transition element that does not show variable oxidation states.**

- scandium ($Z = 21$)

### Intext question

**Q3** Which 3d-series transition metal shows the most oxidation states, and why?

# 4.8 Trends in the $M^{2+}/M$ Standard Electrode Potentials

### Data and comparison

- Table 4.E gives the thermochemical quantities for turning the solid metal into $M^{2+}$ ions in solution, with their standard electrode potentials
- Figure 4.D compares observed $E^{\ominus}$ values with those calculated from these data

### Copper is unique

- its positive $E^{\ominus}$ explains why it cannot release $H_2$ from acids
- only oxidising acids (nitric, hot concentrated sulphuric) attack Cu, and the acids are reduced
- the large energy needed to convert Cu(s) to $Cu^{2+}$(aq) is not made up by its hydration enthalpy

**Figure 4.D** Observed and calculated values for the standard electrode potentials ($M^{2+} \rightarrow M°$) of the elements Ti to Zn

- standard electrode potential (V, from –2 to about +0.4) for Ti, V, Cr, Mn, Fe, Co, Ni, Cu and Zn, with observed and calculated values plotted side by side
- values climb from about –1.6 (Ti) towards the positive value of Cu, with dips at Mn and at Zn; observed and calculated points agree closely for Ti to Fe, and diverge somewhat at Co and Ni

### General trend

- across the series $E^{\ominus}$ generally becomes less negative, which follows the overall increase in (first + second) ionisation enthalpy
- notably, Mn, Ni and Zn have more negative $E^{\ominus}$ than the trend predicts

### Intext question

**Q4** Copper's $E^{\ominus}(M^{2+}/M)$ is positive (+0.34 V). Suggest a reason. (Hint: think about its high $\Delta_a H^{\ominus}$ and low $\Delta_{hyd} H^{\ominus}$.)

### Worked example two $d^4$ ions

**$Cr^{2+}$ and $Mn^{3+}$ are both $d^4$. Why is $Cr^{2+}$ a reducing agent and $Mn^{3+}$ an oxidising agent?**

- $Cr^{2+}$ reduces because it goes from $d^4$ to $d^3$, and $d^3$ has a half-filled $t_{2g}$ level (see Unit 5)
- $Mn^{3+}$ oxidises because becoming $Mn^{2+}$ gives the half-filled $d^5$ configuration, which is extra stable

TABLE 4.E Thermochemical data (kJ mol⁻¹) for the first row Transition Elements and the Standard Electrode Potentials for the Reduction of $M^{II}$ to M.

| Element (M) | $\Delta_a H^\ominus$(M) | $\Delta_i H_1^\ominus$ | $\Delta_i H_2^\ominus$ | $\Delta_{hyd} H^\ominus(M^{2+})$ | $E^\ominus$/V |
|---|---|---|---|---|---|
| Ti | 469 | 656 | 1309 | −1866 | −1.63 |
| V | 515 | 650 | 1414 | −1895 | −1.18 |
| Cr | 398 | 653 | 1592 | −1925 | −0.90 |
| Mn | 279 | 717 | 1509 | −1862 | −1.18 |
| Fe | 418 | 762 | 1561 | −1998 | −0.44 |
| Co | 427 | 758 | 1644 | −2079 | −0.28 |
| Ni | 431 | 736 | 1752 | −2121 | −0.25 |
| Cu | 339 | 745 | 1958 | −2121 | 0.34 |
| Zn | 130 | 906 | 1734 | −2059 | −0.76 |

The half-filled d subshell of $Mn^{2+}$ and the full $d^{10}$ of $Zn^{2+}$ account for their $E^\ominus$ values; Ni's $E^\ominus$ reflects its most negative $\Delta_{hyd}H^\ominus$.

## 4.9 Trends in the $M^{3+}/M^{2+}$ Standard Electrode Potentials

- $E^\ominus(M^{3+}/M^{2+})$ values (Table 4.C) vary irregularly
- Sc: low, as $Sc^{3+}$ has a stable noble gas configuration; Zn: highest, as an electron must come out of stable $d^{10}$ $Zn^{2+}$
- Mn: fairly high, so $Mn^{2+}$ ($d^5$) is especially stable; Fe: fairly low, showing extra stability of $Fe^{3+}$ ($d^5$)
- V: fairly low, reflecting the stability of $V^{2+}$ (half-filled $t_{2g}$ level, Unit 5)

## 4.10 Trends in Stability of Higher Oxidation States

- stable halides of the 3d metals: Table 4.F; highest oxidation numbers in $TiX_4$, $VF_5$ and $CrF_6$
- Mn(+7) has no simple halide, but $MnO_3F$ exists; after Mn only $FeX_3$ and $CoF_3$ are trihalides
- fluorine stabilises the top state through higher lattice energy ($CoF_3$) or higher bond enthalpies in higher covalent compounds ($VF_5$, $CrF_6$)
- V(+5) appears only as $VF_5$; the other halides hydrolyse to oxohalides $VOX_3$; fluorides are also unstable in low oxidation states, e.g. $VX_2$ (X = Cl, Br or I)

TABLE 4.F Formulas of Halides of 3d Metals

| Oxidation number | Ti | V | Cr | Mn | Fe | Co | Ni | Cu | Zn |
|---|---|---|---|---|---|---|---|---|---|
| +6 | | | $CrF_6$ | | | | | | |
| +5 | | $VF_5$ | $CrF_5$ | | | | | | |
| +4 | $TiX_4$ | $VX^{I}_4$ | $CrX_4$ | $MnF_4$ | | | | | |
| +3 | $TiX_3$ | $VX_3$ | $CrX_3$ | $MnF_3$ | $FeX^{I}_3$ | $CoF_3$ | | | |
| +2 | $TiX_2^{III}$ | $VX_2$ | $CrX_2$ | $MnX_2$ | $FeX_2$ | $CoX_2$ | $NiX_2$ | $CuX_2^{II}$ | $ZnX_2$ |
| +1 | | | | | | | | $CuX^{III}$ | |

Key: X = F → I; $X^{I}$ = F → Br; $X^{II}$ = F, Cl; $X^{III}$ = Cl → I

### Copper halides and Cu(I)

- CuX is likewise unstable; every $Cu^{II}$ halide is known except the iodide, because $Cu^{2+}$ oxidises $I^-$ to $I_2$:

$$2Cu^{2+} + 4I^- \rightarrow Cu_2I_2(s) + I_2$$

- many Cu(I) compounds are unstable in water and disproportionate: $2Cu^+ \rightarrow Cu^{2+} + Cu$
- $Cu^{2+}(aq)$ beats $Cu^+(aq)$ in stability because its far more negative $\Delta_{hyd}H^{\ominus}$ more than pays for copper's second ionisation enthalpy

### Oxygen stabilises the highest states

- in the oxides (Table 4.G) the top oxidation number equals the group number, reached from $Sc_2O_3$ to $Mn_2O_7$
- beyond group 7 no iron oxide above $Fe_2O_3$ is known; ferrates(VI), $(FeO_4)^{2-}$, form in alkali but readily break down to $Fe_2O_3$ and $O_2$
- oxocations also stabilise $V^V$ as $VO_2^+$, $V^{IV}$ as $VO^{2+}$ and $Ti^{IV}$ as $TiO^{2+}$

### Oxygen outdoes fluorine

- the highest Mn fluoride is $MnF_4$, but the highest oxide is $Mn_2O_7$
- oxygen's edge comes from its ability to form multiple bonds to metals
- in covalent $Mn_2O_7$ each Mn is tetrahedrally surrounded by O atoms, with an Mn–O–Mn bridge; tetrahedral $[MO_4]^{n-}$ ions are known for $V^V$, $Cr^{VI}$, $Mn^V$, $Mn^{VI}$ and $Mn^{VII}$

### Intext question

**Q5** Explain the irregular variation of the first and second ionisation enthalpies across the first transition series.

TABLE 4.G **Oxides of 3d Metals**

| Oxidation number | Groups | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| +7 | | | | | $Mn_2O_7$ | | | | | |
| +6 | | | | $CrO_3$ | | | | | | |
| +5 | | | $V_2O_5$ | | | | | | | |
| +4 | | $TiO_2$ | $V_2O_4$ | $CrO_2$ | $MnO_2$ | | | | | |
| +3 | $Sc_2O_3$ | $Ti_2O_3$ | $V_2O_3$ | $Cr_2O_3$ | $Mn_2O_3$ $Mn_3O_4$* | $Fe_2O_3$ $Fe_3O_4$* | $Co_3O_4$* | | | |
| +2 | | TiO | VO | (CrO) | MnO | FeO | CoO | NiO | CuO | ZnO |
| +1 | | | | | | | | | $Cu_2O$ | |

* mixed oxides

### Worked example oxidising power of oxoions

Why does oxidising power increase in the order $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$?

- because the lower species to which each is reduced becomes increasingly stable along the series

# 4.11 Chemical Reactivity and $E^{\ominus}$ Values

### Reactive and noble

- chemical reactivity of transition metals varies widely; many are electropositive enough to dissolve in mineral acids, while a few are 'noble', untouched by single acids
- first-series metals except copper are fairly reactive and are oxidised by 1M $H^+$, though the actual rate of reaction with oxidants such as $H^+$ can be slow
- e.g. titanium and vanadium are in practice passive to dilute non-oxidising acids at room temperature

### $M^{2+}/M$ values

- $E^{\ominus}(M^{2+}/M)$ (Table 4.C) show divalent cations forming less readily from left to right; the values become less negative as (first + second) ionisation enthalpy grows
- Mn, Ni and Zn are more negative than the trend suggests: stable $d^5$ in $Mn^{2+}$ and $d^{10}$ in zinc explain theirs; for nickel, its most negative hydration enthalpy

### $M^{3+}/M^{2+}$ couples

- from the $E^{\ominus}(M^{3+}/M^{2+})$ values (Table 4.C), $Mn^{3+}$ and $Co^{3+}$ are the strongest oxidising agents in aqueous solution
- $Ti^{2+}$, $V^{2+}$ and $Cr^{2+}$ are strong reducing agents that liberate hydrogen from dilute acid, e.g. $2Cr^{2+}(aq) + 2H^+(aq) \rightarrow 2Cr^{3+}(aq) + H_2(g)$

### Worked example irregular $E^{\ominus}(M^{2+}/M)$

Explain the irregularity in these $E^{\ominus}(M^{2+}/M)$ values for first-row metals:

| | V | Cr | Mn | Fe | Co | Ni | Cu |
|---|---|---|---|---|---|---|---|
| $E^{\ominus}(M^{2+}/M)$ / V | –1.18 | –0.91 | –1.18 | –0.44 | –0.28 | –0.25 | +0.34 |

- the irregular variation of the ionisation enthalpies ($\Delta_i H_1 + \Delta_i H_2$), together with sublimation enthalpies that are comparatively much smaller for manganese and vanadium

### Worked example $Mn^{3+}/Mn^{2+}$

Why is $E^{\ominus}$ for $Mn^{3+}/Mn^{2+}$ far more positive than for $Cr^{3+}/Cr^{2+}$ or $Fe^{3+}/Fe^{2+}$?

- mainly Mn's much larger third ionisation energy (the change needed is $d^5$ to $d^4$); this is also why the +3 state of Mn matters little

### Intext questions

**Q6** Why does a metal show its highest oxidation state only in its oxide or fluoride?

**Q7** Which is the stronger reducing agent, $Cr^{2+}$ or $Fe^{2+}$? Give the reason.

# 4.12 Magnetic Properties

In an applied magnetic field two main kinds of behaviour appear: **diamagnetism**, where the substance is repelled by the field, and **paramagnetism**, where it is attracted.

### Ferromagnetism and paramagnetism

- substances attracted very strongly are **ferromagnetic**; ferromagnetism is an extreme form of paramagnetism
- many transition metal ions are paramagnetic
- paramagnetism comes from unpaired electrons, each unpaired electron carrying a magnetic moment that arises from its spin and orbital angular momenta

### Spin-only formula

- in first-series compounds the orbital contribution is effectively quenched and does not count; the moment depends on the number of unpaired electrons:

$$\mu = \sqrt{n(n+2)}$$

- $n$ = number of unpaired electrons; μ in Bohr magnetons (BM); one unpaired electron gives 1.73 BM

### Using measured moments

- the moment rises as the number of unpaired electrons rises, so a measured moment is a good guide to how many unpaired electrons an atom, molecule or ion has
- Table 4.H compares spin-only and experimental values for some first-row ions; the experimental data are mostly for hydrated ions in solution or in the solid

Table 4.H **Calculated and Observed Magnetic Moments (BM)**

| Ion | Configuration | Unpaired electron(s) | Magnetic moment | |
|---|---|---|---|---|
| | | | Calculated | Observed |
| $Sc^{3+}$ | $3d^0$ | 0 | 0 | 0 |
| $Ti^{3+}$ | $3d^1$ | 1 | 1.73 | 1.75 |
| $Ti^{2+}$ | $3d^2$ | 2 | 2.84 | 2.76 |
| $V^{2+}$ | $3d^3$ | 3 | 3.87 | 3.86 |
| $Cr^{2+}$ | $3d^4$ | 4 | 4.90 | 4.80 |
| $Mn^{2+}$ | $3d^5$ | 5 | 5.92 | 5.96 |
| $Fe^{2+}$ | $3d^6$ | 4 | 4.90 | 5.3 – 5.5 |
| $Co^{2+}$ | $3d^7$ | 3 | 3.87 | 4.4 – 5.2 |
| $Ni^{2+}$ | $3d^8$ | 2 | 2.84 | 2.9 – 3.4 |
| $Cu^{2+}$ | $3d^9$ | 1 | 1.73 | 1.8 – 2.2 |
| $Zn^{2+}$ | $3d^{10}$ | 0 | 0 | |

### Worked example spin-only magnetic moment

Find the magnetic moment of a divalent ion in aqueous solution whose atomic number is 25.

- the divalent ion of element 25 in solution is $d^5$, with five unpaired electrons:

$$\mu = \sqrt{5(5+2)} = \mathbf{5.92\ BM}$$

### *Intext question*

**Q8** Work out the 'spin only' magnetic moment of the $M^{2+}(aq)$ ion for which $Z = 27$.

## 4.13 Formation of Coloured Ions

### *Why the ions are coloured*

- exciting an electron from a lower-energy d orbital to a higher-energy d orbital absorbs light whose frequency matches the excitation energy (Unit 5); this usually falls in the visible region
- the colour seen is the complement of the light absorbed
- which frequency is absorbed depends on the ligand; in aqueous solution, with water as ligand, the ion colours are those in Table 4.I

**Figure 4.E** Colours of some of the first row transition metal ions in aqueous solutions. From left to right: $V^{4+}$,$V^{3+}$,$Mn^{2+}$,$Fe^{3+}$,$Co^{2+}$,$Ni^{2+}$and $Cu^{2+}$ .

- a row of seven aqueous solutions of these ions, each showing the characteristic colour listed for that ion in Table 4.I

TABLE 4.I *Colours of Some of the First Row (aquated) Transition Metal Ions*

| Configuration | Example | Colour | Configuration | Example | Colour |
|---|---|---|---|---|---|
| $3d^0$ | $Sc^{3+}$ | colourless | $3d^4$ | $Cr^{2+}$ | blue |
| $3d^0$ | $Ti^{4+}$ | colourless | $3d^5$ | $Mn^{2+}$ | pink |
| $3d^1$ | $Ti^{3+}$ | purple | $3d^5$ | $Fe^{3+}$ | yellow |
| $3d^1$ | $V^{4+}$ | blue | $3d^6$ | $Fe^{2+}$ | green |
| $3d^2$ | $V^{3+}$ | green | $3d^6$ / $3d^7$ | $Co^{3+}$ / $Co^{2+}$ | blue / pink |
| $3d^3$ | $V^{2+}$ | violet | $3d^8$ | $Ni^{2+}$ | green |
| $3d^3$ | $Cr^{3+}$ | violet | $3d^9$ | $Cu^{2+}$ | blue |
| $3d^4$ | $Mn^{3+}$ | violet | $3d^{10}$ | $Zn^{2+}$ | colourless |

## 4.14 Formation of Complex Compounds

- **complex compounds**: metal ions attached to a number of anions or neutral molecules, forming complex species that have their own typical properties
- examples: $[Fe(CN)_6]^{3-}$, $[Fe(CN)_6]^{4-}$, $[Cu(NH_3)_4]^{2+}$ and $[PtCl_4]^{2-}$

### Why transition metals form so many complexes

- the chemistry of complexes is treated fully in Unit 5
- the metal ions are comparatively small and highly charged, and have d orbitals available for bonding

## 4.15 Catalytic Properties

### Sources of catalytic activity

- transition metals and their compounds are well known catalysts, because they can take up several oxidation states and form complexes
- examples: vanadium(V) oxide (Contact Process), finely divided iron (Haber's Process), nickel (catalytic hydrogenation)

### Surface catalysis

- bonds form between reactant molecules and atoms on the catalyst surface; first-row metals use 3d and 4s electrons for this
- this concentrates the reactants at the surface and weakens bonds in the reacting molecules, lowering the activation energy

### Catalysis through changing oxidation state

- the ability of transition metal ions to change oxidation state makes them more effective catalysts; e.g. iron(III) catalyses the iodide–persulphate reaction:

$$2I^- + S_2O_8^{2-} \rightarrow I_2 + 2SO_4^{2-}$$

$$2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$$
$$2Fe^{2+} + S_2O_8^{2-} \rightarrow 2Fe^{3+} + 2SO_4^{2-}$$

## 4.16 Formation of Interstitial Compounds

### What they are

- formed when small atoms such as H, C or N are trapped within metal crystal lattices
- usually non-stoichiometric, neither typically ionic nor covalent: TiC, $Mn_4N$, $Fe_3H$, $VH_{0.56}$, $TiH_{1.7}$, etc.; the formulas match no normal oxidation state of the metal
- named **interstitial compounds** after the nature of their composition

### Their characteristics

- (i) melting points higher than those of the pure metals
- (ii) very hard: some borides are nearly as hard as diamond
- (iii) metallic conductivity is retained; (iv) chemically inert

## 4.17 Alloy Formation

### Alloys

- an **alloy** is a blend of metals made by mixing the components; it may be a homogeneous solid solution with one metal's atoms scattered at random among the other's
- such alloys form between atoms whose metallic radii are within about 15 percent of each other; transition metals, alike in radius and other features, alloy readily

### Examples

- the alloys are hard and often high-melting; best known are ferrous alloys: Cr, V, W, Mo and Mn go into a variety of steels and stainless steel
- transition metals alloyed with non-transition metals matter to industry too: brass (copper–zinc), bronze (copper–tin)

### Intext question

**Q9** Why is the $Cu^+$ ion unstable in aqueous solution?

### Worked example disproportionation

**What is 'disproportionation' of an oxidation state? Give an example.**

- an oxidation state disproportionates when it becomes less stable than two others, one lower and one higher
- in acid, manganese(VI) is unstable relative to manganese(VII) and manganese(IV):

$$3Mn^{VI}O_4^{2-} + 4H^+ \rightarrow 2Mn^{VII}O_4^- + Mn^{IV}O_2 + 2H_2O$$

## 4.18 Some Important Compounds of Transition Elements

## 4.19 Oxides and Oxoanions of Metals

### Oxides

- usually made by reacting the metals with oxygen at high temperatures; every metal except scandium forms an ionic MO oxide
- the top oxidation number in the oxides equals the group number, reached from $Sc_2O_3$ to $Mn_2O_7$; beyond group 7 no iron oxide above $Fe_2O_3$ is known
- oxocations also stabilise $V^V$ as $VO_2^+$, $V^{IV}$ as $VO^{2+}$ and $Ti^{IV}$ as $TiO^{2+}$

### Ionic to covalent, basic to acidic

- ionic character falls as the metal's oxidation number rises: $Mn_2O_7$ is a covalent green oil; even $CrO_3$ and $V_2O_5$ melt at low temperatures; these higher oxides are chiefly acidic
- $Mn_2O_7$ gives $HMnO_4$; $CrO_3$ gives $H_2CrO_4$ and $H_2Cr_2O_7$

### Vanadium and chromium oxides

- $V_2O_5$ is amphoteric though mainly acidic, giving $VO_4^{3-}$ and $VO_2^+$ salts
- vanadium shifts steadily from basic $V_2O_3$ to less basic $V_2O_4$ to amphoteric $V_2O_5$; $V_2O_4$ dissolves in acids to give $VO^{2+}$ salts; $V_2O_5$ reacts with alkalies and with acids to give $VO_4^{3-}$ and $VO_2^+$ respectively
- well-characterised CrO is basic, but $Cr_2O_3$ is amphoteric

### Potassium dichromate, $K_2Cr_2O_7$

- a very important chemical in the leather industry and an oxidant for making many azo compounds
- dichromates come from chromates, which are obtained by fusing chromite ore ($FeCr_2O_4$) with sodium or potassium carbonate with free access of air

### Preparation, step by step

$$4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$$

- the yellow sodium chromate solution is filtered and acidified with sulphuric acid; orange sodium dichromate, $Na_2Cr_2O_7.2H_2O$, crystallises from it:

$$2Na_2CrO_4 + 2H^+ \rightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$$

### From sodium to potassium dichromate

- sodium dichromate is the more soluble, so potassium dichromate is made by adding potassium chloride to sodium dichromate solution; orange $K_2Cr_2O_7$ crystallises:

$$Na_2Cr_2O_7 + 2KCl \rightarrow K_2Cr_2O_7 + 2NaCl$$

### Chromate ⇌ dichromate

- interconvert in water according to pH; Cr has the same oxidation state in both

$$2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$$

$$Cr_2O_7^{2-} + 2OH^- \rightarrow 2CrO_4^{2-} + H_2O$$

### Structures

- chromate, $CrO_4^{2-}$: **tetrahedral**
- dichromate, $Cr_2O_7^{2-}$: two tetrahedra sharing one corner, Cr–O–Cr angle 126° (Figure 4.F)

**Figure 4.F**

- **$CrO_4^{2-}$**: Cr at the centre bonded to four O atoms at the corners of a tetrahedron
- **$Cr_2O_7^{2-}$**: two $CrO_4$ tetrahedra joined through one shared O; bridging Cr–O 179 pm, terminal Cr–O 163 pm, Cr–O–Cr angle 126°

### Dichromate as oxidant

- both dichromates oxidise strongly; the sodium salt, being more soluble in water, is the one widely used for oxidations in organic chemistry; potassium dichromate is a primary standard in volumetric analysis
- in acid: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ ($E^\ominus = 1.33$ V)
- so acidified dichromate oxidises iodides to iodine, sulphides to sulphur, tin(II) to tin(IV) and iron(II) salts to iron(III):

$2I^- \rightarrow I_2 + 2e^-$ ; $Sn^{2+} \rightarrow Sn^{4+} + 2e^-$ $\qquad$ $H_2S \rightarrow 2H^+ + S + 2e^-$ ; $Fe^{2+} \rightarrow Fe^{3+} + e^-$

- adding the dichromate half-reaction to that of the reducing agent gives the full ionic equation, e.g.:

$$Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$$

### Potassium permanganate, $KMnO_4$: preparation

- fuse $MnO_2$ with an alkali metal hydroxide and an oxidant such as $KNO_3$; this gives dark green $K_2MnO_4$, which disproportionates in neutral or acidic solution to permanganate:

$$2MnO_2 + 4KOH + O_2 \rightarrow 2K_2MnO_4 + 2H_2O \qquad 3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$$

- commercially: alkaline oxidative fusion of $MnO_2$, then electrolytic oxidation of manganate(VI):

$$MnO_2 \xrightarrow{\text{(KOH fusion; air or } KNO_3 \text{ as oxidant)}} MnO_4^{2-} \text{ (manganate ion)} \;;\; MnO_4^{2-} \xrightarrow{\text{(oxidised by electrolysis, alkaline medium)}} MnO_4^- \text{ (permanganate ion)}$$

## Laboratory route and properties

- in the lab, a manganese(II) salt is oxidised to permanganate by peroxodisulphate:

$$2Mn^{2+} + 5S_2O_8^{2-} + 8H_2O \rightarrow 2MnO_4^- + 10SO_4^{2-} + 16H^+$$

- dark purple, almost black crystals, isostructural with $KClO_4$; only modestly soluble (6.4 g/100 g water at 293 K); decomposes on heating at 513 K:

$$2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$$

## Colour, magnetism, structure

- two notable physical properties: its intense colour, and diamagnetism together with weak temperature-dependent paramagnetism; both are explained by molecular orbital theory, beyond present scope
- manganate and permanganate ions are tetrahedral; π-bonding comes from overlap of oxygen p orbitals with manganese d orbitals
- green manganate is paramagnetic (one unpaired electron); permanganate is diamagnetic (no unpaired electron)

## Acidified permanganate as oxidant

- oxidises oxalates to $CO_2$, iron(II) to iron(III), nitrites to nitrates and iodides to free iodine; the reductant half-reactions:

$5C_2O_4^{2-} \rightarrow 10CO_2 + 10e^-$ ; $5Fe^{2+} \rightarrow 5Fe^{3+} + 5e^-$

$5NO_2^- + 5H_2O \rightarrow 5NO_3^- + 10H^+ + 10e^-$ ; $10I^- \rightarrow 5I_2 + 10e^-$

- add the $KMnO_4$ half-reaction to the reductant's, balancing as needed, to get the full reaction

## Three reduction products

$MnO_4^- + e^- \rightarrow MnO_4^{2-}$ ($E^\ominus = +0.56$ V)

$MnO_4^- + 4H^+ + 3e^- \rightarrow MnO_2 + 2H_2O$ ($E^\ominus = +1.69$ V)

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ ($E^\ominus = +1.52$ V)

- clearly the hydrogen ion concentration strongly influences the reaction
- kinetics matters as well as redox potential: at $[H^+] = 1$ permanganate should oxidise water, but this is extremely slow unless Mn(II) ions are present or the temperature is raised

## Important oxidations by $KMnO_4$: 1. in acid solution

- (a) iodine set free from potassium iodide: $10I^- + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5I_2$
- (b) $Fe^{2+}$ (green) converted to $Fe^{3+}$ (yellow): $5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$

### 1. In acid solution (continued)

- (c) oxalate ion or oxalic acid oxidised at 333 K:

$$5C_2O_4^{2-} + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 10CO_2$$

- (d) hydrogen sulphide oxidised, precipitating sulphur:

$$H_2S \rightarrow 2H^+ + S^{2-}$$

$$5S^{2-} + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5S$$

- (e) sulphurous acid or sulphite oxidised to sulphate or sulphuric acid:

$$5SO_3^{2-} + 2MnO_4^- + 6H^+ \rightarrow 2Mn^{2+} + 3H_2O + 5SO_4^{2-}$$

- (f) nitrite oxidised to nitrate:

$$5NO_2^- + 2MnO_4^- + 6H^+ \rightarrow 2Mn^{2+} + 5NO_3^- + 3H_2O$$

### 2. In neutral or faintly alkaline solution

- (a) a notable reaction, iodide to iodate: $2MnO_4^- + H_2O + I^- \rightarrow 2MnO_2 + 2OH^- + IO_3^-$
- (b) thiosulphate oxidised almost quantitatively to sulphate: $8MnO_4^- + 3S_2O_3^{2-} + H_2O \rightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$
- (c) manganous salt oxidised to $MnO_2$, catalysed by zinc sulphate or zinc oxide: $2MnO_4^- + 3Mn^{2+} + 2H_2O \rightarrow 5MnO_2 + 4H^+$

### Caution and uses

- permanganate titrations do not work well in hydrochloric acid, because HCl is oxidised to chlorine
- uses: analytical work; a popular oxidant for organic preparations; and, thanks to its oxidising power, bleaching wool, cotton, silk and other fibres and removing colour from oils

### The inner transition elements (f-block)

- two series: **lanthanoids** (the 14 elements after lanthanum) and **actinoids** (the 14 after actinium)
- lanthanum closely resembles the lanthanoids and is usually included with them (general symbol Ln); actinium is likewise discussed with the actinoids

## 4.20 The Lanthanoids

### Lanthanoids vs actinoids

- lanthanoids resemble each other more closely than members of any ordinary transition series; with a single stable oxidation state, they let us study how small steps in size and nuclear charge affect a run of otherwise alike elements
- actinoid chemistry is much more complicated, partly from their wide range of oxidation states and partly because their radioactivity makes them hard to study; the two series are treated separately
- Table 4.J gives, for lanthanum and each lanthanoid (Ln), the name, symbol, electronic configuration of the atom and some ions, and the atomic and ionic radii

## 4.21 Electronic Configurations

- the atoms share $6s^2$ but have varying occupancy of the 4f level (Table 4.J)
- every tripositive ion, the most stable oxidation state for all lanthanoids, has the form $4f^n$ ($n$ = 1 to 14 as atomic number increases)

## 4.22 Atomic and Ionic Sizes

### The lanthanoid contraction

- from lanthanum to lutetium, atoms and ions become steadily smaller; this is a feature unique to lanthanoid chemistry, with far-reaching consequences for the third transition series
- the fall in atomic radii (from metal structures) is less regular than the fall in $M^{3+}$ radii (Figure 4.G)
- same cause as in an ordinary transition series, imperfect shielding within a sub-shell; but one 4f electron shields another less than one d electron shields another as nuclear charge rises, giving a fairly regular fall in size

**Figure 4.G** Trends in ionic radii of lanthanoids

- ionic radius (pm, about 90 to 110) against atomic number 57 to 71
- the $M^{3+}$ ions, $La^{3+}$ to $Lu^{3+}$, fall along a smooth, steadily descending curve
- the less common $Ce^{4+}$, $Pr^{4+}$ and $Tb^{4+}$ lie below the $M^{3+}$ curve; $Sm^{2+}$, $Eu^{2+}$, $Tm^{2+}$ and $Yb^{2+}$ lie above it

### Consequences

- the cumulative contraction makes third-series radii very close to those of the corresponding second-series members
- Zr (160 pm) and Hf (159 pm), almost identical in radius because of it, occur together in nature and are hard to separate

## 4.23 Oxidation States

### +3 dominant; +2 and +4 occasional

- La(III) and Ln(III) compounds predominate, but +2 and +4 ions sometimes occur in solution or in solids
- the irregularity (like that in ionisation enthalpies) stems chiefly from the special stability of an f subshell that is empty, half full or full
- Pr, Nd, Tb and Dy show +4 too, but only in oxides $MO_2$; samarium behaves much like europium, showing +2 and +3

### Examples

- **$Ce^{IV}$**: favoured by its noble gas configuration, but a strong oxidant returning to +3; $E^{\ominus}(Ce^{4+}/Ce^{3+})$ = +1.74 V suggests it could oxidise water, but the reaction is very slow, so Ce(IV) makes a good analytical reagent
- **$Eu^{2+}$**: formed by losing both s electrons, its $f^7$ configuration explaining its formation; a strong reductant going to +3
- **$Yb^{2+}$** ($f^{14}$) is a reductant; **$Tb^{IV}$** (half-filled f) is an oxidant

Table 4.1 Electronic Configurations and Radii of Lanthanum and Lanthanoids

| Atomic number | Name | Symbol | Electronic configurations* | | | | Radii / pm | |
|---|---|---|---|---|---|---|---|---|
| | | | Ln | $Ln^{2+}$ | $Ln^{3+}$ | $Ln^{4+}$ | Ln | $Ln^{3+}$ |
| 57 | Lanthanum | La | $5d^1 6s^2$ | $5d^1$ | $4f^0$ | – | 187 | 106 |
| 58 | Cerium | Ce | $4f^1 5d^1 6s^2$ | $4f^2$ | $4f^1$ | $4f^0$ | 183 | 103 |
| 59 | Praseodymium | Pr | $4f^3 6s^2$ | $4f^3$ | $4f^2$ | $4f^1$ | 182 | 101 |
| 60 | Neodymium | Nd | $4f^4 6s^2$ | $4f^4$ | $4f^3$ | $4f^2$ | 181 | 99 |
| 61 | Promethium | Pm | $4f^5 6s^2$ | $4f^5$ | $4f^4$ | – | 181 | 98 |
| 62 | Samarium | Sm | $4f^6 6s^2$ | $4f^6$ | $4f^5$ | – | 180 | 96 |
| 63 | Europium | Eu | $4f^7 6s^2$ | $4f^7$ | $4f^6$ | – | 199 | 95 |
| 64 | Gadolinium | Gd | $4f^7 5d^1 6s^2$ | $4f^7 5d^1$ | $4f^7$ | – | 180 | 94 |
| 65 | Terbium | Tb | $4f^9 6s^2$ | $4f^9$ | $4f^8$ | $4f^7$ | 178 | 92 |
| 66 | Dysprosium | Dy | $4f^{10} 6s^2$ | $4f^{10}$ | $4f^9$ | $4f^8$ | 177 | 91 |
| 67 | Holmium | Ho | $4f^{11} 6s^2$ | $4f^{11}$ | $4f^{10}$ | – | 176 | 89 |
| 68 | Erbium | Er | $4f^{12} 6s^2$ | $4f^{12}$ | $4f^{11}$ | – | 175 | 88 |
| 69 | Thulium | Tm | $4f^{13} 6s^2$ | $4f^{13}$ | $4f^{12}$ | – | 174 | 87 |
| 70 | Ytterbium | Yb | $4f^{14} 6s^2$ | $4f^{14}$ | $4f^{13}$ | – | 173 | 86 |
| 71 | Lutetium | Lu | $4f^{14} 5d^1 6s^2$ | $4f^{14} 5d^1$ | $4f^{14}$ | – | – | – |

* only electrons outside the [Xe] core are shown

## 4.24 General Characteristics

### Physical properties

- all are soft, silvery white metals that tarnish quickly in air; hardness rises with atomic number, samarium being steel-hard
- melting points 1000 to 1200 K, but samarium melts at 1623 K; typical metal structures; conduct heat and electricity well
- density and other properties vary smoothly, except for Eu and Yb and sometimes Sm and Tm

### Colour and magnetism

- many $Ln^{3+}$ ions are coloured, as solids and in solution, owing to their f electrons; $La^{3+}$ and $Lu^{3+}$ are colourless, the rest coloured
- absorption bands are narrow, probably because the excitation is within the f level
- all lanthanoid ions are paramagnetic except the $f^0$ type ($La^{3+}$, $Ce^{4+}$) and the $f^{14}$ type ($Yb^{2+}$, $Lu^{3+}$)

### Ionisation enthalpies

- first about 600 kJ $mol^{-1}$, second about 1200 kJ $mol^{-1}$, comparable with calcium
- detailed study of third ionisation enthalpies suggests exchange enthalpy (as in the 3d orbitals of the first transition series) lends some stability to empty, half-filled and full f levels; hence the unusually low third ionisation enthalpies of La, Gd and Lu

### Chemical behaviour

- early members are quite reactive, like calcium; with rising atomic number they behave more like aluminium
- their reactivity is reflected in $E^{\ominus}$ for the half-reaction $Ln^{3+}(aq) + 3e^- \rightarrow Ln(s)$

### Reactions of the lanthanoids

- $E^{\ominus}(Ln^{3+}/Ln)$ lies between –2.2 and –2.4 V, except Eu at –2.0 V: only a small spread
- combine with hydrogen on gentle heating in the gas; heated with carbon give carbides $Ln_3C$, $Ln_2C_3$ and $LnC_2$
- set hydrogen free from dilute acids and burn in halogens, forming halides; form oxides $M_2O_3$ and hydroxides $M(OH)_3$
- the hydroxides are definite compounds, not merely hydrated oxides, and are basic like alkaline earth oxides and hydroxides; general reactions in Figure 4.H

**Figure 4.H** Chemical reactions of the lanthanoids.

- Ln at the centre with its products around it:
- burns in $O_2 \rightarrow Ln_2O_3$; with acids $\rightarrow H_2$; with halogens $\rightarrow LnX_3$; with $H_2O \rightarrow Ln(OH)_3 + H_2$
- with C at 2773 K $\rightarrow LnC_2$; heated with N $\rightarrow$ LnN; heated with S $\rightarrow Ln_2S_3$

### Uses of the lanthanoids

- chief single use: alloy steels for plates and pipes
- **mischmetall**: about 95% lanthanoid metal and 5% iron, with traces of S, C, Ca and Al; much of it goes into Mg-based alloys for bullets, shells and lighter flints
- mixed lanthanoid oxides catalyse petroleum cracking; some single Ln oxides serve as phosphors in television screens and similar fluorescent surfaces

## 4.25 The Actinoids

The actinoids are the fourteen elements from Th to Lr; their names, symbols and some properties are given in Table 4.K.

TABLE 4.K Some Properties of Actinium and Actinoids

| Atomic number | Name | Symbol | Electronic configurations | | | Radii / pm | |
|---|---|---|---|---|---|---|---|
| | | | M | $M^{3+}$ | $M^{4+}$ | $M^{3+}$ | $M^{4+}$ |
| 89 | Actinium | Ac | $6d^17s^2$ | $5f^0$ | – | 111 | – |
| 90 | Thorium | Th | $6d^27s^2$ | $5f^1$ | $5f^0$ | – | 99 |
| 91 | Protactinium | Pa | $5f^26d^17s^2$ | $5f^2$ | $5f^1$ | – | 96 |
| 92 | Uranium | U | $5f^36d^17s^2$ | $5f^3$ | $5f^2$ | 103 | 93 |
| 93 | Neptunium | Np | $5f^46d^17s^2$ | $5f^4$ | $5f^3$ | 101 | 92 |
| 94 | Plutonium | Pu | $5f^67s^2$ | $5f^5$ | $5f^4$ | 100 | 90 |
| 95 | Americium | Am | $5f^77s^2$ | $5f^6$ | $5f^5$ | 99 | 89 |
| 96 | Curium | Cm | $5f^76d^17s^2$ | $5f^7$ | $5f^6$ | 99 | 88 |
| 97 | Berkelium | Bk | $5f^97s^2$ | $5f^8$ | $5f^7$ | 98 | 87 |
| 98 | Californium | Cf | $5f^{10}7s^2$ | $5f^9$ | $5f^8$ | 98 | 86 |
| 99 | Einsteinium | Es | $5f^{11}7s^2$ | $5f^{10}$ | $5f^9$ | – | – |
| 100 | Fermium | Fm | $5f^{12}7s^2$ | $5f^{11}$ | $5f^{10}$ | – | – |
| 101 | Mendelevium | Md | $5f^{13}7s^2$ | $5f^{12}$ | $5f^{11}$ | – | – |
| 102 | Nobelium | No | $5f^{14}7s^2$ | $5f^{13}$ | $5f^{12}$ | – | – |
| 103 | Lawrencium | Lr | $5f^{14}6d^17s^2$ | $5f^{14}$ | $5f^{13}$ | – | – |

### Radioactivity

- all actinoids are radioactive; the earlier ones have fairly long half-lives, the later ones from a day down to 3 minutes for lawrencium ($Z = 103$)
- the later members could be made only in nanogram amounts, which makes them harder to study

## 4.26 Electronic Configurations

### Filling 5f

- all actinoids are thought to have $7s^2$ with variable occupancy of 5f and 6d
- the fourteen electrons are formally added to 5f, though not in thorium ($Z = 90$); from Pa on, 5f fills until it is complete at element 103

### Irregularities and bonding

- as with the lanthanoids, irregularities relate to the stability of $f^0$, $f^7$ and $f^{14}$ occupancies: Am is [Rn] $5f^7 7s^2$ and Cm is [Rn] $5f^7 6d^1 7s^2$
- 5f orbitals share the angular part of the wave-function with 4f, but are less deeply buried, so 5f electrons take part in bonding far more

## 4.27 Ionic Sizes

- the lanthanoid trend is seen here too: atoms and $M^{3+}$ ions shrink gradually across the series, the **actinoid contraction**
- the contraction from one element to the next is, however, larger than for the lanthanoids, because 5f electrons shield poorly

## 4.28 Oxidation States

### A wider range

- partly because the 5f, 6d and 7s levels have comparable energies; known states in Table 4.L
- +3 in general; the first half of the series often shows higher states: the maximum rises from +4 (Th) to +5 (Pa), +6 (U) and +7 (Np), then falls in later elements

### Compared with lanthanoids

- like the lanthanoids, they have more compounds in +3 than in +4; +3 and +4 ions tend to hydrolyse
- the oxidation states are spread so unevenly, and differ so much between early and late members, that reviewing actinoid chemistry by oxidation state is unsatisfactory

Table 4.L **Oxidation States of Actinium and Actinoids**

| Ac | Th | Pa | U | Np | Pu | Am | Cm | Bk | Cf | Es | Fm | Md | No | Lr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 | | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | | | | | | |
| | | 5 | 5 | 5 | 5 | 5 | | | | | | | | |
| | | | 6 | 6 | 6 | 6 | | | | | | | | |
| | | | | 7 | 7 | | | | | | | | | |

## 4.29 General Characteristics and Comparison with Lanthanoids

The actinoid metals all look silvery but adopt a variety of structures, because their metallic radii are far more irregular than those of the lanthanoids.

### Reactivity

- actinoids are very reactive metals, above all in finely divided form
- boiling water gives a mixture of oxide and hydride; most non-metals combine with them at moderate temperatures
- hydrochloric acid attacks all of them; nitric acid affects most only slightly, as protective oxide layers form; alkalies have no action

### Magnetism and ionisation

- magnetic properties are more complex than for lanthanoids; susceptibility tracks the number of unpaired 5f electrons roughly as for lanthanoids, but lanthanoid values are higher
- ionisation enthalpies of early actinoids, though not accurately known, are lower than those of early lanthanoids
- reasonably so: newly occupied 5f orbitals penetrate the inner core less, so 5f electrons are better shielded from the nucleus than 4f electrons; being less firmly held, the outer electrons are available for bonding

### Comparing the two series

- only in the latter half of the actinoid series does behaviour like that of the lanthanoids appear
- yet even early actinoids resemble lanthanoids in being closely alike and in varying gradually in properties that do not involve a change of oxidation state
- each contraction reaches beyond its own series, altering the sizes, and hence the properties, of the elements after it in the same period; the lanthanoid contraction matters more, since the chemistry of the elements after the actinoids is still much less known

### Worked example +4 lanthanoid

**Name a lanthanoid well known for showing the +4 oxidation state.**

- cerium ($Z = 58$)

### Intext question

**Q10** Why is the actinoid contraction larger from one element to the next than the lanthanoid contraction?

## 4.30 Some Applications of d- and f-Block Elements

### Construction, special compounds, coinage

- iron and steels lead all construction materials; they are made by reducing iron oxides, removing impurities, and adding carbon and alloying metals such as Cr, Mn and Ni
- some compounds are made for special purposes: $TiO_2$ for the pigment industry, $MnO_2$ for dry battery cells; the battery industry also needs Zn and Ni/Cd
- Group 11 elements still deserve the name coinage metals, although Ag and Au are now confined to collectors' items

### Coins today

- present-day UK 'copper' coins are copper-coated steel
- UK 'silver' coins are a Cu/Ni alloy

### Industrial catalysts

- $V_2O_5$: oxidation of $SO_2$ in making sulphuric acid
- $TiCl_4$ with $Al(CH_3)_3$: the Ziegler catalysts for making polyethylene (polythene)
- iron: Haber process, ammonia from $N_2/H_2$; nickel: hydrogenation of fats; $PdCl_2$: Wacker process, ethene to ethanal
- nickel complexes help polymerise alkynes and other organic compounds, benzene among them; AgBr: its special light-sensitivity serves the photographic industry

## Summary

### Blocks and series

- d-block: groups 3–12 in the broad middle of the table, with inner d orbitals filling; f-block: set outside at the foot, with 4f and 5f filling
- three well-recognised transition series follow the filling of 3d, 4d and 5d

### Physical properties

- all behave as typical metals: strong in tension, ductile, malleable, good conductors of heat and electricity, metallic in character
- high melting and boiling points from ($n$–1)d electrons joining strong interatomic bonding; maxima near mid-series show one unpaired electron per d orbital favours strong interaction

### Characteristic behaviour

- successive ionisation enthalpies rise less steeply than in main group elements, so losing varying numbers of ($n$–1)d electrons is not energetically unfavourable
- ($n$–1)d involvement brings variable oxidation states, paramagnetism, catalytic activity, and a tendency to form coloured ions, interstitial compounds and complexes
- chemical behaviour varies widely: many dissolve in mineral acids, a few are 'noble'; in the first series all but copper are fairly reactive

### Compounds

- they form binary compounds with non-metals such as oxygen, nitrogen, sulphur and halogens; first-series oxides are usually made from metal and oxygen at high temperature and dissolve in acids and bases to give oxometallic salts
- $K_2Cr_2O_7$: from chromite ore fused with alkali in air, then the extract acidified; $KMnO_4$: from pyrolusite ($MnO_2$); both ions are strong oxidising agents

### f-block

- lanthanoids and actinoids make up the f-block; as 4f fills, atomic and ionic sizes fall steadily (lanthanoid contraction), with far-reaching effects on the elements after them
- lanthanum and the lanthanoids are rather soft white metals that react readily with water to give solutions of +3 ions; +3 is the main state, though some also show +4 and +2

## Summary, continued

- the +4 and +2 states occur only occasionally among the lanthanoids
- actinoid chemistry is more complex because they exist in many oxidation states, and the radioactivity of many of them makes study difficult
- d- and f-block elements and their compounds have many uses, notably in steels, catalysts, complexes and organic syntheses

## Exercises

**E1** Give the electronic configuration of: (i) $Cr^{3+}$ (ii) $Pm^{3+}$ (iii) $Cu^{+}$ (iv) $Ce^{4+}$ (v) $Co^{2+}$ (vi) $Lu^{2+}$ (vii) $Mn^{2+}$ (viii) $Th^{4+}$

**E2** Why do $Mn^{2+}$ compounds resist oxidation to the +3 state better than $Fe^{2+}$ compounds?

**E3** Explain briefly why the +2 state grows steadily more stable with rising atomic number across the first half of the first transition row.

**E4** How far do electronic configurations decide which oxidation states are stable in the first transition series? Give examples.

**E5** What stable oxidation state would you expect for transition elements whose atoms have the ground-state d configurations $3d^3$, $3d^5$, $3d^8$ and $3d^4$?

**E6** Name the first-series oxometal anions in which the metal's oxidation state equals its group number.

**E7** What is the lanthanoid contraction, and what are its consequences?

**E8** What are the characteristics of transition elements, and why are they so called? Which d-block elements might not be counted as transition elements?

**E9** How does the electronic configuration of transition elements differ from that of non-transition elements?

**E10** Which oxidation states do the lanthanoids show?

**E11** Give reasons: (i) transition metals and many of their compounds are paramagnetic; (ii) transition metals have high enthalpies of atomisation; (iii) transition metals usually form coloured compounds; (iv) transition metals and many of their compounds are good catalysts.

**E12** What are interstitial compounds, and why are they so common for transition metals?

**E13** How does the variability of oxidation states in transition metals differ from that in non-transition metals? Give examples.

**E14** Describe how potassium dichromate is made from iron chromite ore. What happens to a potassium dichromate solution as its pH is raised?

**E15** How does potassium dichromate act as an oxidant? Write the ionic equations for its reactions with (i) iodide, (ii) iron(II) solution and (iii) $H_2S$.

## Exercises continued

**E16** Describe how potassium permanganate is made. How does acidified permanganate react with (i) iron(II) ions, (ii) $SO_2$ and (iii) oxalic acid? Give the ionic equations.

**E17** Some $E^{\ominus}$ values for $M^{2+}/M$ and $M^{3+}/M^{2+}$ are: $Cr^{2+}/Cr$ –0.9 V, $Cr^{3+}/Cr^{2+}$ –0.4 V; $Mn^{2+}/Mn$ –1.2 V, $Mn^{3+}/Mn^{2+}$ +1.5 V; $Fe^{2+}/Fe$ –0.4 V, $Fe^{3+}/Fe^{2+}$ +0.8 V. Use them to comment on (i) how stable $Fe^{3+}$ is in acid solution compared with $Cr^{3+}$ or $Mn^{3+}$, and (ii) how readily iron is oxidised compared with chromium or manganese metal.

**E18** Which of these are coloured in aqueous solution: $Ti^{3+}$, $V^{3+}$, $Cu^{+}$, $Sc^{3+}$, $Mn^{2+}$, $Fe^{3+}$ and $Co^{2+}$? Give a reason for each.

**E19** Compare how stable the +2 oxidation state is across the first transition series.

**E20** Compare actinoid and lanthanoid chemistry, especially regarding (i) electronic configuration, (ii) atomic and ionic sizes, (iii) oxidation state and (iv) chemical reactivity.

**E21** Account for these: (i) among $d^4$ species, $Cr^{2+}$ is strongly reducing but manganese(III) strongly oxidising; (ii) cobalt(II) is stable in water but easily oxidised when complexing reagents are present; (iii) the $d^1$ configuration is very unstable in ions.

**E22** What does ‘disproportionation’ mean? Give two examples of disproportionation in aqueous solution.

**E23** Which first-series transition metal shows the +1 state most often, and why?

**E24** Work out the number of unpaired electrons in the gaseous ions $Mn^{3+}$, $Cr^{3+}$, $V^{3+}$ and $Ti^{3+}$. Which is most stable in aqueous solution?

**E25** Give examples of, and suggest reasons for: (i) a transition metal’s lowest oxide is basic, its highest amphoteric/acidic; (ii) oxides and fluorides are where a transition metal shows its top oxidation state; (iii) a metal’s oxoanions carry its highest oxidation state.

**E26** List the steps by which (i) chromite ore is turned into $K_2Cr_2O_7$, and (ii) pyrolusite ore into $KMnO_4$.

**E27** What are alloys? Name an important alloy containing some lanthanoid metals, and give its uses.

**E28** What are inner transition elements? Which of these atomic numbers belong to inner transition elements: 29, 59, 74, 95, 102, 104?

**E29** Actinoid chemistry is less smooth than lanthanoid chemistry. Support this with examples from their oxidation states.

**E30** Which element ends the actinoid series? Give its electronic configuration and comment on its likely oxidation state.

## Exercises continued

**E31** Using Hund's rule, work out the electronic configuration of the $Ce^{3+}$ ion and its 'spin-only' magnetic moment.

**E32** Name the lanthanoids that show the +4 state and those that show +2. Relate this behaviour to their electronic configurations.

**E33** Set actinoid chemistry beside lanthanoid chemistry, looking at (i) how the electrons are arranged, (ii) which oxidation states occur and (iii) how reactive the elements are.

**E34** Give the electronic configurations of the elements with atomic numbers 61, 91, 101 and 109.

**E35** Compare the first transition series with the second and third series in the same vertical columns, paying particular attention to (i) electron arrangement, (ii) oxidation states, (iii) ionisation enthalpies and (iv) sizes of atoms.

**E36** State the number of 3d electrons in $Ti^{2+}$, $V^{2+}$, $Cr^{3+}$, $Mn^{2+}$, $Fe^{2+}$, $Fe^{3+}$, $Co^{2+}$, $Ni^{2+}$ and $Cu^{2+}$, and show how you would expect the five 3d orbitals to be filled in these hydrated (octahedral) ions.

**E37** Discuss the claim that first-series transition elements have many properties unlike those of the heavier transition elements.

**E38** What do these magnetic moments tell you about the complex species?

| Example | Magnetic moment (BM) |
|---|---|
| $K_4[Mn(CN)_6]$ | 2.2 |
| $[Fe(H_2O)_6]^{2+}$ | 5.3 |
| $K_2[MnCl_4]$ | 5.9 |

## Answers to some intext questions

| Question | Answer |
|---|---|
| Q1 | silver ($Z$ = 47) can show the +2 state, in which its d orbitals (4d) are incompletely filled; hence it is a transition element |
| Q2 | zinc uses no 3d electrons in metallic bonding, whereas every other 3d-series metal always uses d electrons in its metallic bonds |
| Q3 | manganese ($Z$ = 25), since its atom has the most unpaired electrons |
| Q5 | mainly the differing stability of different 3d configurations ($d^0$, $d^5$, $d^{10}$ are exceptionally stable) |
| Q6 | oxygen and fluorine, small and very electronegative, are able to take the metal up to its top oxidation state |
| Q7 | $Cr^{2+}$ is the stronger reducing agent: $Cr^{2+} \rightarrow Cr^{3+}$ is $d^4 \rightarrow d^3$, while $Fe^{2+} \rightarrow Fe^{3+}$ is $d^6 \rightarrow d^5$; in a medium such as water $d^3$ is more stable than $d^5$ (see CFSE) |
| Q9 | $Cu^+$ disproportionates in water, $2Cu^+(aq) \rightarrow Cu^{2+}(aq) + Cu(s)$, for which $E^0$ is favourable |
| Q10 | 5f electrons are more effectively shielded from the nuclear charge; in other words, the 5f electrons themselves shield poorly from one element to the next |